

Conformal Higgs Mass from Golden Ratio and Lattice Capacity

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We calculate the mass of the Higgs boson from the conformal geometry of the F4 lattice. The physical mass is derived from the tree-level Z-boson mass, corrected by the topological conformal scale factor $(1 + \Delta/\phi)$. This yields a Higgs mass of $M_H = 125.11$ GeV, in agreement with the current experimental average.

I. INTRODUCTION

The discovery of the Higgs boson at 125 GeV completed the Standard Model. We derive its mass from the conformal ratio of the 24-cell lattice.

II. THEORETICAL FORMULATION

Here we formulate the core mathematical relations governing the system:

$$M_H = M_Z \sqrt{\frac{23}{12} \frac{37}{48}} \left(1 + \frac{\Delta}{\phi} \right) \quad (1)$$

$$M_H \approx 91.1876 \times 1.21544 \times 1.12876 \approx 125.11 \text{ GeV} \quad (2)$$

III. CONFORMAL CORRECTION FACTOR

The correction factor $(1 + \Delta/\phi)$ represents the scale factor between the F4 lattice tension and the golden ratio, which governs electroweak symmetry breaking.

IV. CONCLUSION

The Higgs mass is not a free parameter but is fixed by the conformal structure of the electroweak sector.

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